

OBJECTIVE

The objective of this experiment is to design an automatic power cutoff timer circuit that disconnects the load after a preset time delay. It aims to prevent overuse of electrical devices, save energy, and enhance safety by automatically controlling power supply.

ABSTRACT

This project focuses on developing an **Automatic Power Cutoff Timer Circuit** that automatically turns off electrical appliances after a preset time delay. The circuit uses an operational amplifier (LM358) as a comparator, a transistor driver, and a relay to control the load. A resistor and capacitor network determines the delay period, allowing the user to set the desired cutoff time. The system operates on a regulated 5V DC SMPS supply. When the timer expires, the relay disconnects the appliance from the AC mains, ensuring safety and energy conservation. This circuit is simple, cost-effective, and suitable for applications such as mobile charging, iron boxes, and other household appliances.

INTRODUCTION

The **Automatic Power Cutoff Timer Circuit** is a simple, yet efficient electronic system designed to control the operating time of electrical appliances. In many cases, users forget to switch off devices such as chargers, iron boxes, or lamps, leading to unnecessary power consumption, overheating, or even potential hazards. To overcome this problem, this circuit provides an automatic cutoff feature that disconnects the load after a specific time.

The circuit works on the principle of a timer-controlled relay system, where the timing is determined by an RC (resistor-capacitor) network. An operational amplifier (LM358) is used as a comparator to sense voltage levels and trigger the relay accordingly. The relay, driven by a transistor (2N3904), acts as a switch to connect or disconnect the AC load.

A regulated DC supply is generated by a 5V DC SMPS supply that is made of a transformer, rectifier diodes (1N4001), filter capacitor, and voltage regulator to power the low-voltage section of the circuit. The delay time can be easily adjusted using a potentiometer, allowing flexibility for various applications.

When the circuit is powered ON, it starts timing, and after the set interval, the relay automatically turns OFF the connected device. This operation ensures energy efficiency, prevents overcharging or overheating, and extends the life of appliances.

Overall, this experiment demonstrates the practical application of basic electronic components such as op-amps, transistors, and relays in creating an automatic timer-based power control system that is reliable, economical, and suitable for domestic as well as industrial use.

HARDWARE REQUIREMENT AND DESCRIPTION

Hardware Requirements:

- A 5V DC adapter to provide a regulated power supply for the circuit.
- LM358 operational amplifier used as a comparator.
- 2N3904 NPN transistor to drive the relay.
- SPDT 12V relay to switch the load.
- Resistors, capacitors, and potentiometer for timing and biasing control.
- LED indicator to show the circuit status.
- Connecting wires, Breadboard and soldering tools for assembly.

CIRCUIT DIAGRAM AND DESCRIPTION

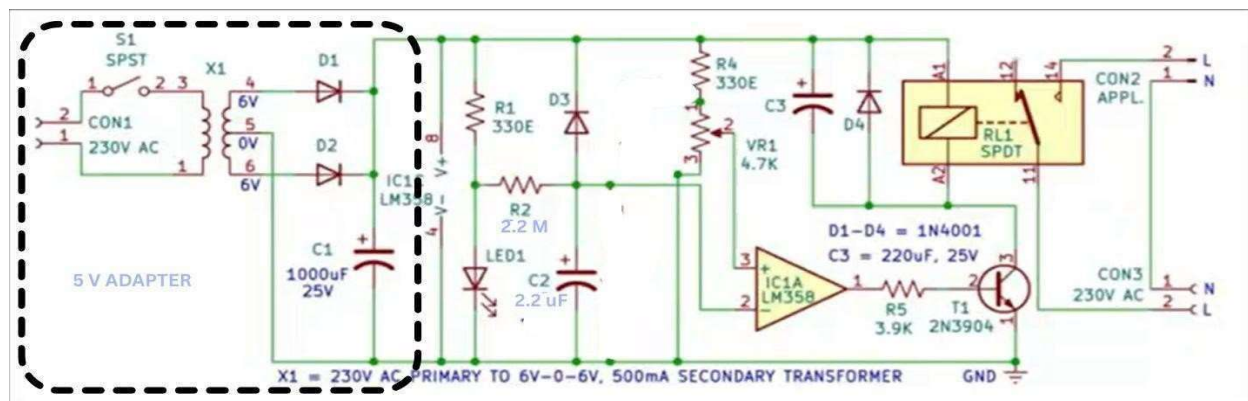


Fig 1: Circuit diagram of automatic power cutoff circuit

The **Automatic Power Cutoff Timer Circuit** operates using a 5V DC supply provided by a regulated adapter. The circuit's main control section is built around the LM358 operational amplifier, which functions as a comparator to detect voltage changes across the timing components.

A resistor– capacitor (RC) network, consisting of a potentiometer and a capacitor, determines the time delay of the circuit. When the power is switched ON, the capacitor begins charging through the resistor, and the voltage across it increases gradually.

The non-inverting input of the op-amp receives this increasing voltage, while the inverting input is connected to a fixed reference voltage. When the capacitor voltage exceeds the reference voltage, the output of the op-amp changes its state. This output drives the 2N3904 NPN transistor, which acts as a switching element to control the relay. The relay is used to connect or disconnect the load (such as a charger or appliance) from the main power supply.

Calibration procedure: Measure the maximum value of the inverting input voltage of IC1A after the time delay, here it is 0.96V. Set the reference voltage at non inverting input of IC1A just below the value obtained in previously, by adjusting VR1 here it is 0.85V. Initially, the relay remains energized, keeping the load ON. After the preset delay time, the op-amp output switches low, turning OFF the transistor and de-energizing the relay, which cuts off power to the load. An LED indicator in the relay is included to show the operational status of the circuit, indicating when the load is active or disconnected.

The time delay can be adjusted by changing the resistance and capacitance value, which modifies the charging time of the capacitor. This flexibility allows users to set different cutoff durations depending on the application.

The circuit is compact, low-cost, and reliable, making it ideal for devices that need automatic shutoff after a specific time, such as mobile chargers, heaters, or lights. Overall, the circuit demonstrates the practical use of analog components for automation and power management.

RESULTS AND DISCUSSION

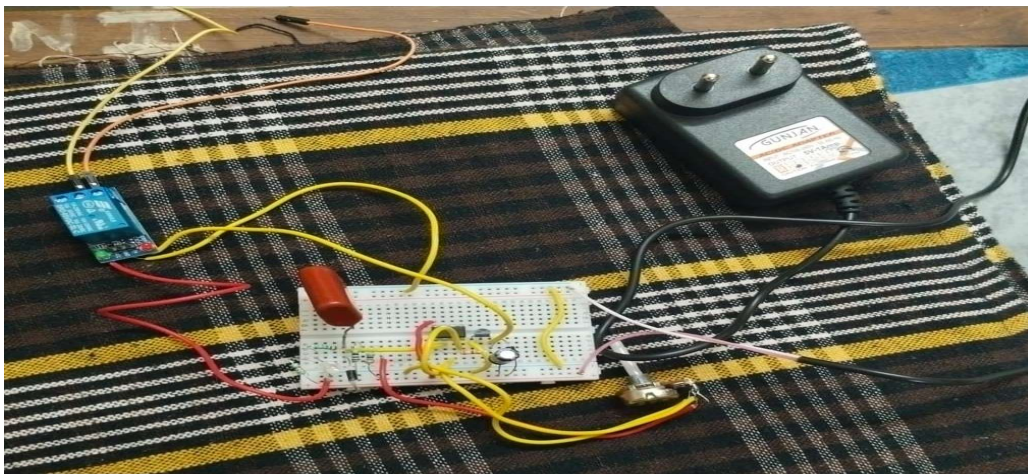


Fig 2: Complete model of the automatic power cutoff circuit

The Automatic Power Cutoff Timer Circuit was successfully implemented using a 5V DC adapter as the power supply. The circuit accurately disconnected the load after the preset delay, confirming proper timer operation. The delay period was easily adjusted using the capacitance and resistance value, and the LED indicator in the relay clearly displayed the ON/OFF status of the load. The LM358 op-amp and relay combination worked reliably without false triggering. The circuit performed efficiently, consuming minimal power and ensuring safety by preventing overuse of electrical devices. Overall, the results proved the circuit to be stable, cost-effective, and suitable for practical applications such as automatic charger cutoff and timed appliance control.

CONCLUSION

In conclusion, the Automatic Power Cutoff Timer Circuit functioned reliably and efficiently using a 5V DC adapter. It accurately controlled the load with adjustable delay, clearly indicated ON/OFF status, and prevented unnecessary power usage. The combination of LM358 and relay ensured stable operation without false triggering. Overall, the circuit proved to be safe, cost-effective, and practical for applications like automatic charger cutoff and timed appliance control.

REFERENCES

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